

Permutations and Combinations

Section 6.3

Section Summary

- Permutations
- Combinations
- Combinatorial Proofs

Permutations

Definition: A *permutation* of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of r elements of a set is called an *r -permutation*.

Example: Let $S = \{1, 2, 3\}$.

- The ordered arrangement 3,1,2 is a permutation of S .
- The ordered arrangement 3,2 is a 2-permutation of S .
- The number of r -permutations of a set with n elements is denoted by $P(n, r)$.
 - The 2-permutations of $S = \{1, 2, 3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3, 2) = 6$.

A Formula for the Number of Permutations

Theorem 1: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$$

r -permutations of a set with n distinct elements.

Proof: Use the product rule. The first element can be chosen in n ways. The second in $n-1$ ways, and so on until there are $(n-(r-1))$ ways to choose the last element.

- Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary 1: If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Solving Counting Problems by Counting Permutations

Example: How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution:

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970,200$$

Solving Counting Problems by Counting Permutations (*continued*)

Example: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

Solving Counting Problems by Counting Permutations (*continued*)

Example: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Combinations

Definition: An *r*-combination of elements of a set is an unordered selection of *r* elements from the set. Thus, an *r*-combination is simply a subset of the set with *r* elements.

- The number of *r*-combinations of a set with *n* distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*. (We will see the notation again in the binomial theorem in Section 6.4.)

Example: Let *S* be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from *S*. It is the same as $\{d, c, a\}$ since the order listed does not matter.

- $C(4, 2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations

Theorem 2: The number of r -combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n, r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n, r) \cdot P(r, r)$.
Therefore,

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations

Example: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$\begin{aligned} C(52, 5) &= \frac{52!}{5!47!} \\ &= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960 \end{aligned}$$

- The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960.$$

This is a special case of a general result. →

Combinations

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

and

$$C(n, n - r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!}.$$

Hence, $C(n, r) = C(n, n - r)$. ◀

This result can be proved without using algebraic manipulation. →

Combinatorial Proofs

- **Definition 1:** A *combinatorial proof* of an identity is a proof that uses one of the following methods.
 - A *double counting proof* uses counting arguments to prove that both sides of an identity count the same objects, but in different ways.
 - A *bijective proof* shows that there is a bijection between the sets of objects counted by the two sides of the identity.

Combinatorial Proofs

- Here are two combinatorial proofs that

$$C(n, r) = C(n, n - r)$$

when r and n are nonnegative integers with $r < n$:

- Bijjective Proof:* Suppose that S is a set with n elements. The function that maps a subset A of S to $\bar{A}$ is a bijection between the subsets of S with r elements and the subsets with $n - r$ elements. Since there is a bijection between the two sets, they must have the same number of elements. ◀
- Double Counting Proof:* By definition the number of subsets of S with r elements is $C(n, r)$. Each subset A of S can also be described by specifying which elements are not in A , i.e., those which are in $\bar{A}$. Since the complement of a subset of S with r elements has $n - r$ elements, there are also $C(n, n - r)$ subsets of S with r elements. ◀

Combinations

Example: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

Solution: By Theorem 2, the number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution: By Theorem 2, the number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775.$$

【Example】 A soccer club has 8 female and 7 male members. For today's match, how many possible configurations are there?

- (1) The coach wants to have 6 female and 5 male players on the grass.
- (2) The coach wants to have 11 players with at most 5 male players on the grass.

Solution:

$$\begin{aligned}
 (1) \quad & C(8, 6) \cdot C(7, 5) \\
 &= 8!/(6! \cdot 2!) \cdot 7!/(5! \cdot 2!) \\
 &= 28 \cdot 21 \\
 &= 588
 \end{aligned}$$

$$(2) \quad C(8, 6)C(7, 5) + C(8, 7)C(7, 4) + C(8, 8)C(7, 3)$$

15

Homework

Sec 6.3: 20, 44, 46

Binomial Coefficients and Identities

Section 6.4

Section Summary

- The Binomial Theorem
- Pascal's Identity and Triangle
- Other Identities Involving Binomial Coefficients

Powers of Binomial Expressions

Definition: A *binomial* expression is the sum of two terms, such as $x + y$. (More generally, these terms can be products of constants and variables.)

- We can use counting principles to find the coefficients in the expansion of $(x + y)^n$ where n is a positive integer.
- To illustrate this idea, we first look at the process of expanding $(x + y)^3$.
- $(x + y)(x + y)(x + y)$ expands into a sum of terms that are the product of a term from each of the three sums.
- Terms of the form x^3, x^2y, xy^2, y^3 arise. The question is what are the coefficients?
 - To obtain x^3 , an x must be chosen from each of the sums. There is only one way to do this. So, the coefficient of x^3 is 1.
 - To obtain x^2y , an x must be chosen from two of the sums and a y from the other. There are $\binom{3}{2}$ ways to do this and so the coefficient of x^2y is 3.
 - To obtain xy^2 , an x must be chosen from one of the sums and a y from the other two. There are $\binom{3}{1}$ ways to do this and so the coefficient of xy^2 is 3.
 - To obtain y^3 , a y must be chosen from each of the sums. There is only one way to do this. So, the coefficient of y^3 is 1.
- We have used a counting argument to show that $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$.
- Next we present the binomial theorem gives the coefficients of the terms in the expansion of $(x + y)^n$.

Binomial Theorem

Binomial Theorem: Let x and y be variables, and n a nonnegative integer. Then:

$$(x+y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

Proof: We use combinatorial reasoning. The terms in the expansion of $(x + y)^n$ are of the form $x^{n-j}y^j$ for $j = 0, 1, 2, \dots, n$. To form the term $x^{n-j}y^j$, it is necessary to choose $n-j$ x s from the n sums. Therefore, the coefficient of $x^{n-j}y^j$ is $\binom{n}{n-j}$ which equals $\binom{n}{j}$. ◀

Using the Binomial Theorem

Example: What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?

Solution: We view the expression as $(2x + (-3y))^{25}$.
By the binomial theorem

$$(2x + (-3y))^{25} = \sum_{j=0}^{25} \binom{25}{j} (2x)^{25-j} (-3y)^j.$$

Consequently, the coefficient of $x^{12}y^{13}$ in the expansion is obtained when $j = 13$.

$$\binom{25}{13} 2^{12} (-3)^{13} = -\frac{25!}{13!12!} 2^{12} 3^{13}.$$

A Useful Identity

Corollary 1: With $n \geq 0$, $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Proof (using binomial theorem): With $x = 1$ and $y = 1$, from the binomial theorem we see that:

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{(n-k)} = \sum_{k=0}^n \binom{n}{k}. \quad \blacktriangleleft$$

Proof (combinatorial): Consider the subsets of a set with n elements. There are $\binom{n}{0}$ subsets with zero elements, $\binom{n}{1}$ with one element, $\binom{n}{2}$ with two elements, ..., and $\binom{n}{n}$ with n elements. Therefore the total is

$$\sum_{k=0}^n \binom{n}{k}.$$

Since, we know that a set with n elements has 2^n subsets, we conclude:

$$\sum_{k=0}^n \binom{n}{k} = 2^n. \quad \blacktriangleleft$$

【 Corollary 2 】 Let n be a positive integer. Then

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

Proof:

Using the Binomial Theorem with $x = 1$ and $y = -1$.

Remark:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots$$

23

Blaise Pascal
(1623-1662)



Pascal's Identity

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof (combinatorial): Let T be a set where $|T| = n + 1$, $a \in T$, and $S = T - \{a\}$. There are $\binom{n+1}{k}$ subsets of T containing k elements. Each of these subsets either:

- contains a with $k - 1$ other elements, or
- contains k elements of S and not a .

There are

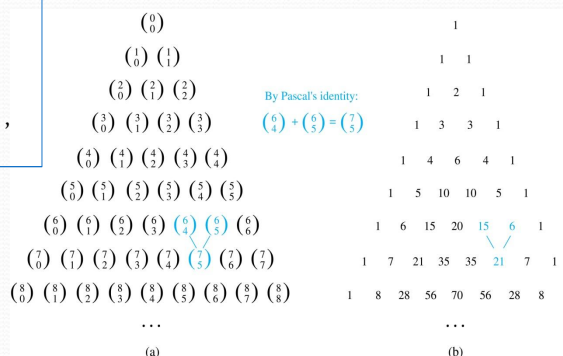
- $\binom{n}{k-1}$ subsets of k elements that contain a , since there are $\binom{n}{k-1}$ subsets of $k - 1$ elements of S ,
- $\binom{n}{k}$ subsets of k elements of T that do not contain a , because there are $\binom{n}{k}$ subsets of k elements of S .

Hence, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$ ◀

See Exercise 19
for an algebraic
proof.

Pascal's Triangle

The n th row in the triangle consists of the binomial coefficients $\binom{n}{k}$, $k = 0, 1, \dots, n$.



By Pascal's identity, adding two adjacent binomial coefficients results in the binomial coefficient in the next row between these two coefficients.

【 Theorem 3 】 *Vandermonde's Identity*

Let m, n and r be nonnegative integers with r not exceeding either m or n . Then

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}$$

Proof:

A and B are two disjoint sets. $|A|=m$, $|B|=n$,

$C(m+n, r)$ ---- the number of ways to pick r elements from $A \cup B$

Another way to pick r elements from $A \cup B$ is to pick $r-k$ elements from A and then k elements from B , where $0 \leq k \leq r$, which can be done in $C(m, r-k) C(n, k)$

【 Corollary 4 】 If n is a nonnegative integer. Then

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2$$

Proof:

We use Vandermonde's Identity with $m = r = n$ to obtain

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{n-k} \binom{n}{k} = \sum_{k=0}^n \binom{n}{k}^2$$

27

【 Theorem 4 】 Let n and r be nonnegative integer with $r \leq n$. Then

$$\binom{n+1}{r+1} = \sum_{j=r}^n \binom{j}{r}$$

Proof:

The left-hand side counts the bit strings of length $n+1$ containing $r+1$ 1s.

We show that the right-hand side counts the same objects by considering the cases corresponding to the possible locations of the final 1 in a string with $r+1$ ones.

$$\sum_{k=r+1}^{n+1} \binom{k-1}{r} = \sum_{j=r}^n \binom{j}{r}$$

28

Homework

Sec.6.4: 18, 26, 30, 34

Generalized Permutations and Combinations

Section 6.5

Section Summary

- Permutations with Repetition
- Combinations with Repetition
- Permutations with Indistinguishable Objects
- Distributing Objects into Boxes

Permutations with Repetition

Theorem 1: The number of r -permutations of a set of n objects with repetition allowed is n^r .

Proof: There are n ways to select an element of the set for each of the r positions in the r -permutation when repetition is allowed. Hence, by the product rule there are n^r r -permutations with repetition. ◀

Example: How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r , which is the number of r -permutations of a set with 26 elements.

Combinations with Repetition

Example: How many ways are there to select five bills from a box containing at least five of each of the following denominations: \$1, \$2, \$5, \$10, \$20, \$50, and \$100?

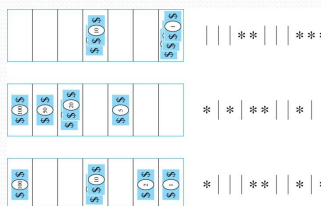
Solution: Place the selected bills in the appropriate position of a cash box illustrated below:



continued →

Combinations with Repetition

- Some possible ways of placing the five bills:



- The number of ways to select five bills corresponds to the number of ways to arrange six bars and five stars in a row.
- This is the number of unordered selections of 5 objects from a set of 11. Hence, there are

$$C(11, 5) = \frac{11!}{5!6!} = 462$$

ways to choose five bills with seven types of bills.

Combinations with Repetition

Theorem 2: The number of r -combinations from a set with n elements when repetition of elements is allowed is

$$C(n + r - 1, r) = C(n + r - 1, n - 1).$$

Proof: Each r -combination of a set with n elements with repetition allowed can be represented by a list of $n - 1$ bars and r stars. The bars mark the n cells containing a star for each time the i th element of the set occurs in the combination.

The number of such lists is $C(n + r - 1, r)$, because each list is a choice of the r positions to place the stars, from the total of $n + r - 1$ positions to place the stars and the bars. This is also equal to $C(n + r - 1, n - 1)$, which is the number of ways to place the $n - 1$ bars. ◀



Combinations with Repetition

Example: Suppose that a cookie shop has four different kinds of cookies. How many different ways can six cookies be chosen?

Solution: The number of ways to choose six cookies is the number of 6-combinations of a set with four elements. By Theorem 2

$$C(9, 6) = C(9, 3) = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$$

is the number of ways to choose six cookies from the four kinds.

Combinations with Repetition

Example: How many solutions does the equation

$$x_1 + x_2 + x_3 = 11$$

have, where x_1 , x_2 and x_3 are nonnegative integers?

Solution: Each solution corresponds to a way to select 11 items from a set with three elements; x_1 elements of type one, x_2 of type two, and x_3 of type three.

By Theorem 2 it follows that there are

$$C(3 + 11 - 1, 11) = C(13, 11) = C(13, 2) = \frac{13 \cdot 12}{1 \cdot 2} = 78$$

solutions.

【Example】 How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 16$$

where $x_i (i=1,2,3,4)$ is nonnegative integer?

Solution:

Since a solution of this equation corresponds to a way of selecting 16 items from a set with four elements so that x_1 items of type one, x_2 items of type two, x_3 items of type three, x_4 items of type four are chosen.

Hence the number of solutions is

$$\begin{aligned} C(4-1+16, 16) &= C(19, 16) \\ &= C(19, 3) \end{aligned}$$

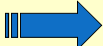
6.5 Generalized Permutations and Combinations

【Example】 How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 16$$

Where $x_i (i = 1, 2, 3, 4)$ is nonnegative integer?

Question:

(1) $x_i > 1$, for $i = 1, 2, 3, 4$  $x_i \geq 2$

$$C(4-1+8, 8) = C(11, 8) = C(11, 3)$$

(2) $x_1 + x_2 + x_3 + x_4 \leq 16$

We can introduce an auxiliary variable x_5 so that

$$x_1 + x_2 + x_3 + x_4 + x_5 = 16$$

$$C(5-1+16, 16) = C(20, 16) = C(20, 4)$$

39

Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

TABLE 1 Combinations and Permutations With and Without Repetition.

Type	Repetition Allowed?	Formula
r -permutations	No	$\frac{n!}{(n-r)!}$
r -combinations	No	$\frac{n!}{r!(n-r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420.$$

The reasoning can be generalized to the following theorem. →

Permutations with Indistinguishable Objects

Theorem 3: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, ..., and n_k indistinguishable objects of type k , is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

Proof: By the product rule the total number of permutations is:

$C(n, n_1) C(n - n_1, n_2) \cdots C(n - n_1 - n_2 - \cdots - n_k, n_k)$ since:

- The n_1 objects of type one can be placed in the n positions in $C(n, n_1)$ ways, leaving $n - n_1$ positions.
- Then the n_2 objects of type two can be placed in the $n - n_1$ positions in $C(n - n_1, n_2)$ ways, leaving $n - n_1 - n_2$ positions.
- Continue in this fashion, until n_k objects of type k are placed in $C(n - n_1 - n_2 - \cdots - n_k, n_k)$ ways.

The product can be manipulated into the desired result as follows:

$$\frac{n!}{n_1!(n - n_1)!} \frac{(n - n_1)!}{n_2!(n - n_1 - n_2)!} \cdots \frac{(n - n_1 - \cdots - n_{k-1})!}{n_k!0!} = \frac{n!}{n_1!n_2!\cdots n_k!}.$$

【Example】 There are 50 students in a class.

- (1) How many ways to select 7 students to construct a leading group?

$$C(50,7)$$

- (2) If two of the leading group are elected as a monitor and a vice monitor, then how many ways are there?

$$= \frac{P(50,7)}{1!1!5!}$$

- (3) If these 7 students are elected to have different tasks, then how many are there?

$$P(50,7)$$

43

【Example】

- (1) How many bit strings of length 10?
- (2) How many bit strings of length 10 are there that contain exactly two 0s, eight 1s?

$$2^{10}$$

$$10!/(2!8!)$$

【Example 3】 How many different strings can be made from the letters in MISSISSIPPI, using all the letters?

Solution:

$$A = \{1, M, 4, I, 4, S, 2, P\}$$

$$\frac{11!}{4!4!2!}$$

44

Distributing Objects into Boxes

- Many counting problems can be solved by counting the ways objects can be placed in boxes.
 - The objects may be either different from each other (*distinguishable*) or identical (*indistinguishable*).
 - The boxes may be labeled (*distinguishable*) or unlabeled (*indistinguishable*).

Distributing Objects into Boxes

- *Distinguishable objects and distinguishable boxes.*
 - There are $n!/(n_1!n_2!\cdots n_k!)$ ways to distribute n distinguishable objects into k distinguishable boxes.
 - (See Exercises 47 and 48 for two different proofs.)
 - Example: There are $52!/(5!5!5!5!32!)$ ways to distribute hands of 5 cards each to four players.

Distributing Objects into Boxes

- *Indistinguishable objects and distinguishable boxes.*
 - There are $C(n + r - 1, n - 1)$ ways to place r indistinguishable objects into n distinguishable boxes.
 - Proof based on one-to-one correspondence between n -combinations from a set with k -elements when repetition is allowed and the ways to place n indistinguishable objects into k distinguishable boxes.
 - Example: There are $C(8 + 10 - 1, 10) = C(17, 10) = 19,448$ ways to place 10 indistinguishable objects into 8 distinguishable boxes.

Distributing Objects into Boxes

- *Distinguishable objects and indistinguishable boxes.*
 - Example: There are 14 ways to put four employees into three indistinguishable offices

[[Example]] How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

Solution:

We represent the four employees by A,B,C,D.

(1) All four are put into one office: **1 ways** {A,B,C,D}

(2) Three are put into one office and a fourth is put into a second office:

4 ways {{A,B,C},{D}}; {{A,B,D},{C}}; {{A,C,D},{B}}; {{B,C,D},{A}}

(3) Two are put into one office and two put into a second office:

3 ways {{A,B},{C,D}}; {{A,D},{B,C}}; {{A,C},{B,D}}

(4) Two are put into one office, and one each put into the other two office:

6 ways

{{A,B},{C},{D}}; {{A,C},{B},{D}}; {{A,D},{B},{C}}; {{B,C},{A},{D}};
{{B,D},{A},{C}}; {{C,D},{A},{B}}

49

[[Example]] How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

Solution:

There are **1+4+3+6=14** ways to put four different employees into three indistinguishable offices.

Another way:

Look at the number of offices into which we put employees.

- 1) There are **6** ways to put four different employees into **three** indistinguishable offices so that no office is empty.
- 2) There are **7** ways to put four different employees into **two** indistinguishable offices so that no office is empty.
- 3) There are **1** ways to put four different employees into **one** offices so that it is not empty.

50

Distributing Objects into Boxes

- *Distinguishable objects and indistinguishable boxes.*
 - There is no simple closed formula for the number of ways to distribute n distinguishable objects into j indistinguishable boxes.
 - See the text for a formula involving *Stirling numbers of the second kind*.

Distributing Objects into Boxes

- *Indistinguishable objects and indistinguishable boxes.*

⌈Example ⌋ How many ways are there to pack six copies of the same book into four identical boxes, where a box can contain as many as six books?

Solution:

We can enumerate all ways to pack the books. For each ways to pack the books, we will list the number of books in the box with the largest of books, followed by the number of books in each box containing at least one book, in order of decreasing number of books in a box.

The ways we can pack the books are

6	5,1	
4,2	4,1,1	
3,3	3,2,1	
3,1,1,1	2,2,2	2,2,1,1

Distributing Objects into Boxes

- *Indistinguishable objects and indistinguishable boxes.*
 - The number of ways of distributing n indistinguishable objects into k indistinguishable boxes equals $p_k(n)$, the number of ways to write n as the sum of at most k positive integers in increasing order.
 - No simple closed formula exists for this number.

Homework

第八版 Sec. 6.5 10, 16, 28, 34, 48, 52, 56, 63

Generating Permutations and Combinations

Section 6.6

Section Summary

- Generating Permutations
- Generating Combinations

Generating Permutations

Problem:

List the permutations of any set of n elements.

Solution:

- Any set with n elements can be placed in one-to-one correspondence with the set $\{1, 2, \dots, n\}$
- Generate the permutation of the n smallest positive integers, and then replace these integers with the corresponding elements.

Generating Permutations

The lexicographic ordering of the set of permutations of $\{1, 2, \dots, n\}$

The permutation $a_1a_2\dots a_n$ precedes the permutation of $b_1b_2\dots b_n$, if for some k , with $1 \leq k \leq n$, $a_1 = b_1, a_2 = b_2, \dots, a_{k-1} = b_{k-1}$, and $a_k < b_k$.

For example:

123465 124635

Generating Permutations

Algorithm of producing the $n!$ permutations of the integers 1, 2, ..., n

- ❖ Begin with the smallest permutation in lexicographic order, namely $1234\dots n$.
- ❖ Produce the next larger permutation.
- ❖ Continue until all $n!$ permutations have been found.

Generating Permutations

Given permutation $a_1a_2\dots a_n$, find the next larger permutation in increasing order:

(1) Find the integers

a_j, a_{j+1} with $a_j < a_{j+1}$ and $a_{j+1} > a_{j+2} > \dots > a_n$

(2) Put in the j th position the least integer among

$a_{j+1}, a_{j+2}, \dots, a_n$ that is greater than a_j

(3) List in increasing order the rest of the integers

$a_j, a_{j+1}, \dots, a_n$

Generating Permutations

Example 1:

What is the next larger permutation in
lexicographic order after 124653?

Solution:

- The next largest permutation of 124653 in lexicographic order is 125346

Generating Combinations

Problem 1:

Generate all combinations of the elements of a finite set .

Solution:

- A combination is just a subset. $\Rightarrow$ We need to list all subsets of the finite set.
- Use bit strings of length n to represent a subset of a set with n elements. $\Rightarrow$ We need to list all bit strings of length n .
- The 2^n bit strings can be listed in order of their increasing size as integers in their binary expansions.

Generating Combinations

Algorithm of producing all bit strings

- ❖ Start with the bit string $000\dots 00$, with n zeros.
- ❖ Then, successively find the next larger expansion until the bit string $111\dots 11$ is obtained.

The method to find the next larger binary expansion:

Locate the first position from the right that is not a 1, then changing all the 1s to the right of this position to 0s and making this first 0 a 1.

For example:

$1000110011 \rightarrow 1000110100$

Generating Combinations

Problem 1:

Generate all r -combinations of the set $\{1, 2, \dots, n\}$

Solution:

The algorithm for generating the r -combination of the set $\{1, 2, \dots, n\}$

- (1) $S_1 = \{1, 2, \dots, r\}$
- (2) If $S_i = \{a_1, a_2, \dots, a_r\}, 1 \leq i \leq C_n^r - 1$ has found, then the next combination can be obtained using the following rules.

First, locate the last element a_i in the sequence such that $a_i \neq n - r + i$. Then replace a_i with $a_i + 1$ and a_j with $a_i + j - i + 1$, for $j = i + 1, i + 2, \dots, r$.

Generating Permutations

Example 2:

$S_i = \{2, 3, 5, 6, 9, 10\}$ is given from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find S_{i+1} .

Solution:

$$S_{i+1} = \{2, 3, 5, 7, 8, 9\}$$

Homework

Sec. 6.6 6(f), 7, 9